

Inle Myat Bush

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Education

Harvard Medical School

2025 - Present

Ph.D. in Neuroscience.

Cornell University

2021 - 2025

B.A. in Computer Science, B.A. in Mathematics, & B.A. in Biology.

Experience

Han Lab, Cornell University

02/2022 - 05/2025

Research Assistant

- Computational and optogenetic dissection of dendrite morphogenesis
 - Developed 10+ computer vision and data analysis pipelines for analysis of time-lapse images of dendrite development in Java, Python.
 - Coauthored 1 paper and 1 book chapter on methods developed.
- Mechanisms of synapse pruning in *Drosophila* olfactory receptor neurons
 - Designed and led investigation of cellular mechanisms and behavioral implications of excitation induced synapse pruning.

Navlakha Lab, Cold Spring Harbor Laboratory

01/2022 - 05/2025

CSHL NSF REU - Research Intern

- Information processing in network dynamics of neuromuscular junction development.
 - Analyzed connectome data, simulation & analysis of dynamical system model.
 - Coauthored paper on work published in *PNAS*.
- Quantified and modeled network dynamics in *Arabidopsis* root systems
 - Lead quantification of time-lapse videos of *Arabidopsis* root structure dynamics.
 - Formalized plant root foraging in reinforcement learning framework.

Ruta Lab, Rockefeller University

06/2024 - 08/2024

Chemers Neustein Summer Undergraduate Research Fellow

- Neuroethological study of Territorial Aggression in *Drosophila prolongata*
 - Designed a suite of ethological experiments to dissect territorial aggression and courtship.
 - Presented posters at SURF Symposium and Cornell Undergraduate Fall Forum.

Su Lab, Colorado University Boulder

05/2019 - 08/2019

Research Intern

- Full time summer intern in genetics and cancer research, studying tumor culling mechanisms during development in *Drosophila melanogaster*.
- Coauthored paper published in *PLOS Genetics* (Recognized as second author).

CIS Teaching Staff, Cornell University

05/2023 - Present

Teaching Assistant, Machine Learning

- 12hr/week for 1 semesters.

Teaching Assistant, Computational Genetics and Genomics

- 12hr/week for 1 semester. Designed/taught 3 lectures to 125 student class. Mentored student teams on semester-long research projects. Administered homeworks & exams.

Teaching Assistant, Design and Analysis of Algorithms

- 12hr/week for 2 semesters. In person tutoring via office hours and recitation sections. Administered and graded algorithm design problem sets and exams.

Publications

1. Dasgupta S., Meirovitch Y., Zheng X., **Bush, I.**, Lichtman J., Navlakha S. (2024). A neural algorithm for computing bipartite matchings. *Proceedings of the National Academy of Science*. doi:10.1073/pnas.2321032121

2. Xu Y., Wang B. **Bush, I.**, Saunders H.A., Wildonger J., Han C. (2024). In vivo optogenetic manipulations of endogenous proteins reveal spatiotemporal roles of microtubule and kinesin in dendrite patterning. *Science Advances*. doi:10.1126/sciadv.adp0138

3. Xu Y., **Bush, I.**, Han C. (2023). Visualization and Measurement of Dendrite Arborization in Neurons. Chapter in *Neuronal Morphogenesis - Methods and Protocols*. doi:10.1007/978-1-0716-3969-6

4. Brown, J., **Bush, I.**, Bozon, J., & Su, T. T. (2020). Cells with loss-of-heterozygosity after exposure to ionizing radiation in Drosophila are culled by p53-dependent and p53-independent mechanisms. *PLOS Genetics*, 16(10). doi:10.1371/journal.pgen.1009056

Selected Projects

- **State Space Models for CRE Functional Classification** (Cornell CS 4775 Computational Genomics) Implemented and benchmarked SOTA deep learning architectures on genomic classification tasks.
- **RNAFoldml** (Cornell CS 3110 Functional Programming) OCaml package that enables users to input both RNA sequences in FASTA format and a set of constraints to predict RNA secondary structure.
- **Score!:** **Database and API** (Cornell CS 1998 Backend Development) Core contributor to SQL database and Flask (Python) API backend for college sports IOS app.

Technical Skills

Programming Languages: Python, R, Java, C, C++, Ocaml, Julia, Matlab, SQL, HTML/CSS, IJM, Javascript

Libraries/Frameworks: Pytorch, NumPy, SciPy, Pandas, MatplotLib, SkLearn, Flask, Jupyter, Maven, Postman, Git, JUnit, ImageJ, Microsoft Office, Dev. Tools (VS Code, Eclipse, IntelliJ, RStudio)

Experimental Techniques: *Drosophila* husbandry and genetics; immunohistochemistry; tissue dissection; confocal microscopy; PCR; behavioral assays; bioinformatics.

Languages: English, French, Burmese (beginner).

Honors

Dean of Science Graduate Fellowship, MIT (Declined)	2024
Undergraduate Research Fund Grant Recipient, Cornell College of Arts and Science	2024
Chemers Neustein Summer Undergraduate Research Fellow, Rockefeller University	2024
NSF Scholar, Cold Spring Harbor Laboratory Undergraduate Research Program	2023
Einhorn Research Grant Recipient, Cornell College of Arts and Science	2022
Best Overall Prize, Cornell AppDev Hack Challenge	2021
Sponsor Prize, Cornell BRH Hackathon	2021
Dean’s List, Cornell College of Arts and Sciences	all semesters

Selected Coursework

Biology:	Computer Science:	Mathematics:
Neuroscience	Machine Learning	Numerical Analysis of Diff. Eq.
Cell Biology	Analysis of Algorithms	Honors Analysis
Genetics	Matrix Computations	Probability
Computational Genetics	Backend Development	Computational Algebra
Ecology	Operating Systems	Linear Algebra
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